Einstein's General Theory of Relativity

Lecture 6: Local Flatness, Parallel Transport, and Curvature

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Chapter 1

Local Flatness, Parallel Transport, and Curvature

Overview. The equivalence principle lets us remove gravity at one event, but not throughout a region containing tidal forces. We will make that statement precise in two stages. First we use coordinate freedom to set the first derivatives of the metric, and hence the connection, to zero at one point. Then we transport vectors around closed curves and discover the obstruction to extending that local flattening: curvature.

1.1 Proper Time in Space-Time

There are two objectives for this lecture. We want to understand geodesics more fully, and we want to connect geodesic motion with gravitational force. Both objectives require us to work in space-time rather than space alone.

Space-time has a metric just as an ordinary geometric space does. Greek indices run over the four coordinates,

$$\mu, \nu = 0, 1, 2, 3, \tag{1.1}$$

with 0 denoting time and $1, 2, 3$ denoting space. Along a timelike worldline, the invariant interval is the proper time:

$$d\tau^2 = g_{\mu\nu}\, dx^\mu dx^\nu. \tag{1.2}$$

In an inertial frame of special relativity and in units $c = 1$,

$$\begin{aligned} \eta_{\mu\nu} &= \mathrm{diag}(1, -1, -1, -1), \\ d\tau^2 &= dt^2 - dx^2 - dy^2 - dz^2. \end{aligned} \tag{1.3}$$

If τ and t are measured in units of time while the spatial coordinates retain units of length, restoring the speed of light gives

$$d\tau^2 = dt^2 - \frac{dx^2 + dy^2 + dz^2}{c^2}. \tag{1.4}$$

For a particle with ordinary speed v,

$$d\tau = dt\sqrt{1 - \frac{v^2}{c^2}}. \tag{1.5}$$

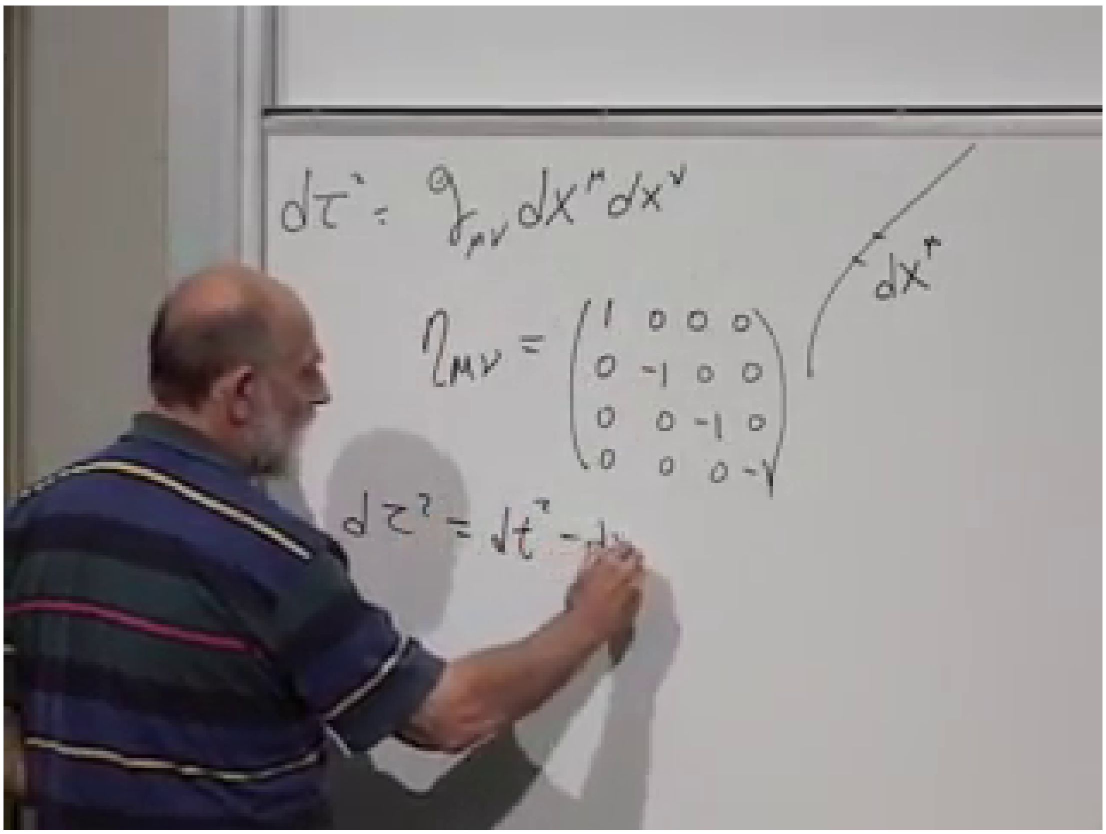

Figure 1.1: The proper-time interval, the Minkowski metric with signature $(+ - - -)$, and the component form of the interval.

Lecture frame: 00:04:00.

When $v \ll c$, proper time and coordinate time are nearly equal. More generally, $d\tau$ is what a clock carried by the particle records. The coordinate t is a label assigned by the chosen frame.

Question from the audience. Does a clock moving with the object record dt, or does it record $d\tau$? [a]

Response. It records $d\tau$. A tape measure laid along a curve records distance along that curve rather than either Cartesian coordinate. In the same way, the moving clock is a space-time tape measure: its ticks mark proper time along the worldline.

[a]Lecture timestamp: 00:06:57–00:08:02.

The formal apparatus is now familiar. The metric, covariant and contravariant tensors, Christoffel symbols, and covariant derivatives are constructed in space-time exactly as in ordinary space. The essential difference is the Lorentzian signature: one direction has the opposite sign from the other three.

Question from the audience. Does enlarging the metric from four dimensions to five or more account for the extra dimensions discussed in string theory? [a]

Response. The dimension is exactly the number of independent coordinates and therefore the size of the metric matrix. This statement is not peculiar to string theory; the tensor rules have the same form in any dimension. A particular theory must still explain why those dimensions are present and what geometry they possess.

[a]Lecture timestamp: 00:09:38–00:10:22.

1.2 Accelerated Coordinates Are Not Yet Curvature

In flat space-time we can always use inertial Cartesian coordinates, so curved coordinates may look like an unnecessary complication. They become useful when the observer is accelerated. Relative to inertial (T, X)-coordinates, the coordinate lines of a uniformly accelerated family are hyperbolic. A freely moving body follows a straight worldline in the inertial chart but appears accelerated in the curved chart.

This is the elevator lesson behind the equivalence principle. An accelerated frame produces an apparent gravitational field. Conversely, inside a freely falling laboratory, a nearly uniform gravitational field can be removed locally. Curved coordinates can therefore imitate gravity. They do not, by themselves, imply that space-time is curved.

The word *locally* is decisive. Near a star, different parts of a large elevator accelerate by slightly different amounts and directions. Those relative accelerations are tidal effects. By shrinking the elevator we can make them arbitrarily small at one event, but no single accelerated coordinate system removes the field throughout an extended region. The mathematical obstruction to doing so is curvature.

1.3 Global Flatness and Local Inertial Coordinates

A space is flat if there exists a coordinate system in which the metric components are constant throughout the region:

$$\partial_r g_{mn} = 0 \qquad \text{everywhere in the region.} \tag{1.6}$$

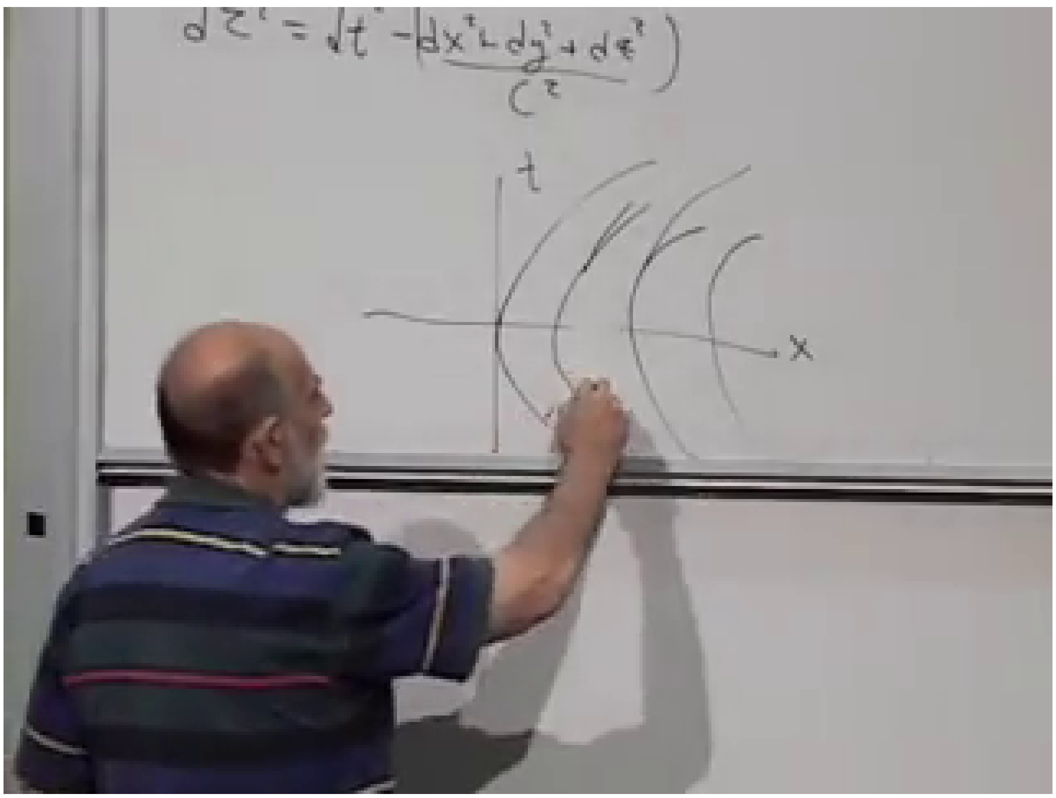

Figure 1.2: Straight inertial axes and a family of curved accelerated coordinate lines. A straight inertial worldline acquires coordinate acceleration in the accelerated description.

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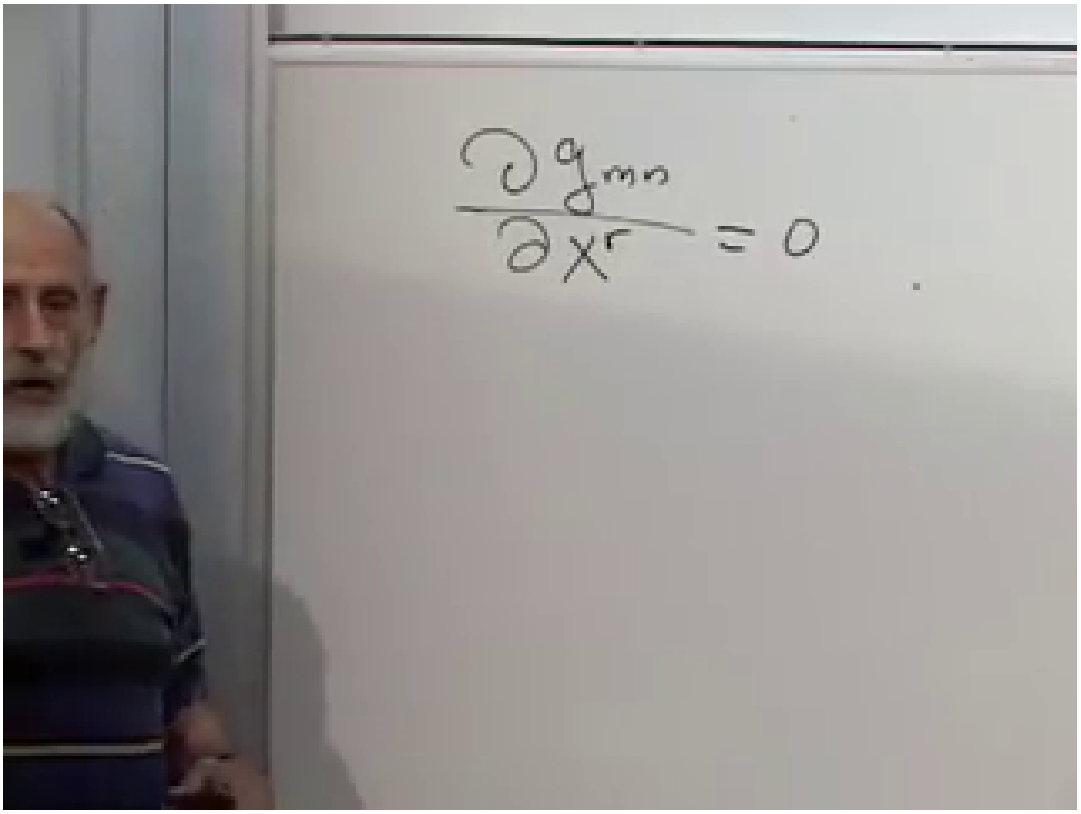

Figure 1.3: Flatness expressed by coordinates in which the first derivatives of the metric vanish everywhere. The same condition can always be achieved at one regular point even when the geometry is curved.

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The constant matrix need not literally be δ_{mn}. Uniform oblique coordinates produce constant off-diagonal entries. A further linear change of basis can put a positive-definite metric into Euclidean form, or a Lorentzian metric into Minkowski form.

Curved geometry does not permit Eq. (1.6) throughout a neighborhood, but smooth coordinate freedom always permits the pointwise conditions

$$g_{mn}(p) = \delta_{mn}, \qquad \partial_r g_{mn}(p) = 0 \tag{1.7}$$

in a Riemannian space. In space-time, δ_{mn} is replaced by $\eta_{\mu\nu}$. These are normal coordinates, or local inertial coordinates in the Lorentzian case. They flatten the metric through first order at p, not across a finite region.

1.4 Counting the Coordinate Freedom

The lecture's local argument begins after a linear transformation has already aligned and normalized the metric at p. Put the common origin at $x = 0 = y$, and consider a quadratic change of coordinates,

$$y^m = x^m + C^m{}_{rs} x^r x^s + O(x^3). \tag{1.8}$$

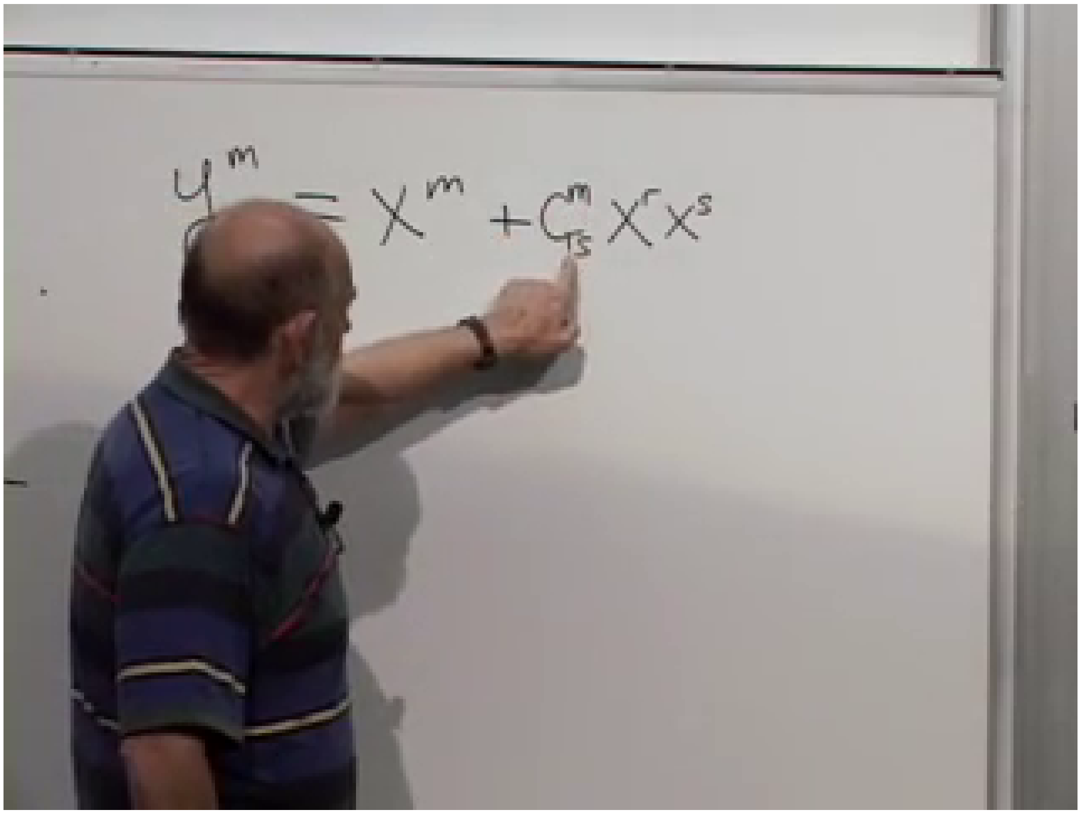

Figure 1.4: The quadratic coordinate change used to adjust first derivatives of the metric at a chosen point.

Lecture frame: 00:22:58.

Only the part of $C^m{}_{rs}$ symmetric in r, s contributes, because $x^r x^s = x^s x^r$. The quadratic term changes second derivatives of the coordinate map and therefore changes first derivatives of the metric at the origin, while leaving the origin and the leading linear basis fixed.

Question from the audience. Has the linear term in the Taylor expansion been dropped? [a]

Response. It is the x^m term in Eq. (1.8). A general invertible linear matrix could stand there, but that freedom has already been used to choose the basis at the point. The quadratic coefficients are the new freedom needed to alter ∂g.

[a] Lecture timestamp: 00:23:44–00:24:16.

In D dimensions, a symmetric pair r, s has

$$\frac{D(D+1)}{2} \tag{1.9}$$

independent values. Since the upper index m has D values, the quadratic transformation contains

$$N_C = D\,\frac{D(D+1)}{2} = \frac{D^2(D+1)}{2} \tag{1.10}$$

coefficients. The target conditions

$$\left.\frac{\partial g_{mn}(y)}{\partial y^s}\right|_p = 0 \tag{1.11}$$

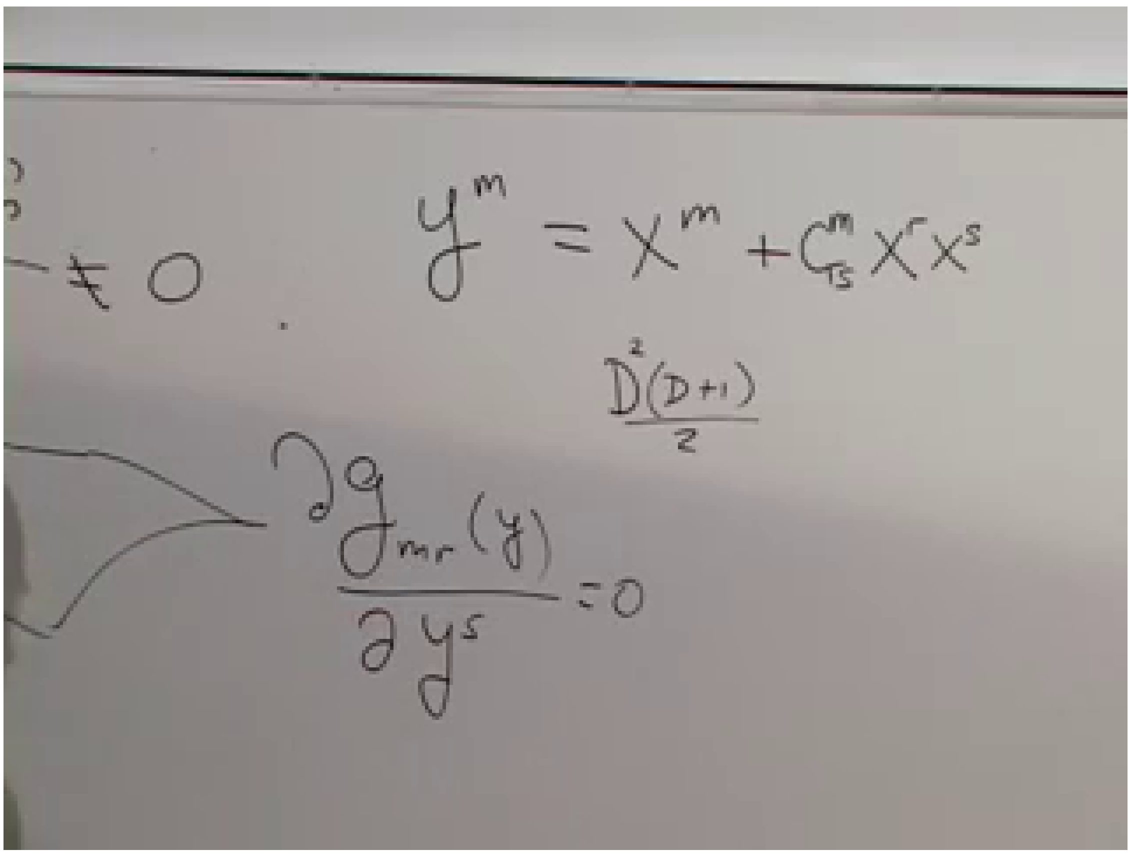

Figure 1.5: The unknowns-versus-equations count. The symmetric quadratic coefficients match the number of first-derivative conditions on the metric.

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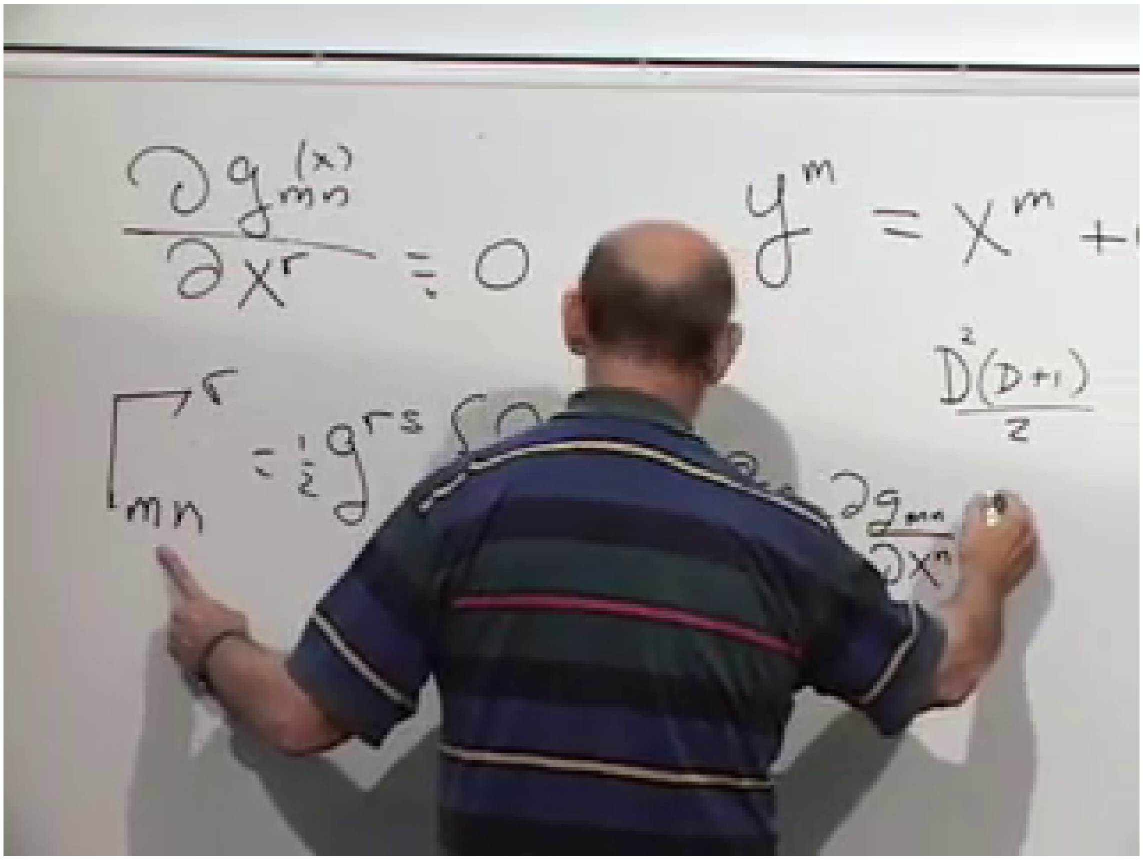

Figure 1.6: The Christoffel formula beside the local condition $\partial g = 0$. Every connection coefficient vanishes at the selected point in normal coordinates.

Lecture frame: 00:30:20.

have exactly the same count: m, n form a symmetric pair and s is independent.

Counting alone warns us what should be possible; the explicit normal-coordinate construction proves that the relevant linear system is solvable for a smooth, nondegenerate metric. The same coordinate freedom does not generically remove every second derivative. Those surviving second-order effects are where curvature first appears.

1.5 The Connection at a Locally Inertial Point

For the torsion-free, metric-compatible connection,

$$\Gamma^r{}_{mn} = \frac{1}{2} g^{rs} \left(\partial_m g_{sn} + \partial_n g_{sm} - \partial_s g_{mn} \right). \tag{1.12}$$

Thus normal coordinates satisfy

$$\Gamma^r{}_{mn}(p) = 0. \tag{1.13}$$

This does not make the connection a tensor. In fact, the ability to make a nonzero set of connection coefficients vanish at one point by changing coordinates is direct evidence that $\Gamma^r{}_{mn}$ is not tensorial. Curvature cannot be removed this way.

The Levi–Civita connection also obeys metric compatibility:

$$\begin{aligned}\nabla_r g_{mn} &= \partial_r g_{mn} - \Gamma^s{}_{rm} g_{sn} - \Gamma^s{}_{rn} g_{ms} \\ &= 0. \end{aligned} \tag{1.14}$$

In normal coordinates at p, both ∂g and Γ vanish, so Eq. (1.14) is immediate there. Because ∇g is a tensor, its vanishing is then coordinate independent.

Question from the audience. Do the quadratic coefficients have to fit together continuously from point to point, and is the argument for $\nabla g = 0$ circular? [a]

Response. A different normal chart may be chosen at each point, and for a smooth metric those local choices can be made smoothly on a sufficiently small patch. They do not combine into one chart with $\Gamma = 0$ everywhere unless the curvature vanishes. The apparent circularity is also real: metric compatibility is part of the definition of the Levi–Civita connection, together with zero torsion. The normal coordinate argument checks that this standard construction is consistent; it is not an independent derivation of every possible connection.

[a] Lecture timestamp: 00:32:58–00:35:01.

1.5.1 Coordinate and geometric singularities

All of this assumes a sufficiently smooth, nondegenerate metric. A bad coordinate chart may make an ordinary point look singular. Polar coordinates fail at the origin because the angle is undefined there, but Cartesian coordinates make the Euclidean metric perfectly regular.

A genuine geometric singularity cannot be repaired by changing coordinates. The tip of an ideal cone is the lecture's first example: the surface is locally flat away from the tip, while no smooth chart turns the apex into an ordinary Euclidean point.

Question from the audience. Could the metric be infinitely wiggly, with derivatives that fail to exist, so that no locally flat coordinates can be found? [a]

Response. Differential geometry assumes the regularity required for the derivatives under discussion. A rough coordinate expression may still be a coordinate artifact if another smooth chart repairs it. If no chart makes the metric regular, the point is genuinely singular and the normal-coordinate theorem does not apply there.

[a] Lecture timestamp: 00:35:02–00:38:18.

Question from the audience. Is solving for the quadratic coefficients the same as finding coordinates in which the Christoffel symbols vanish? [a]

Response. At the selected regular point, yes. Choosing the coefficients so that $\partial g = 0$ makes Eq. (1.12) vanish there. The result is pointwise, not global, and it again shows why a connection coefficient is not itself a tensor.

[a] Lecture timestamp: 00:38:18–00:39:17.

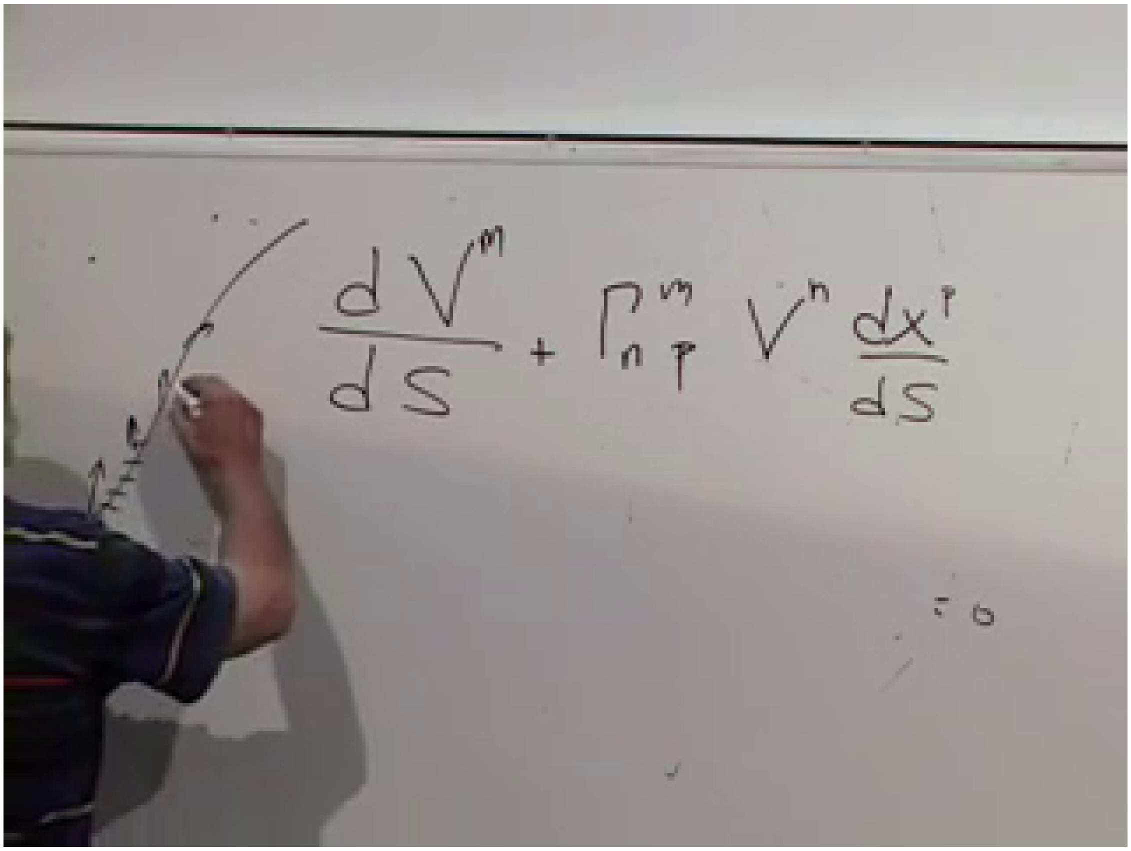

Figure 1.7: The covariant derivative of a vector along a curve: ordinary component change plus the connection correction contracted with the tangent.

Lecture frame: 00:44:12.

1.6 Covariant Differentiation Along a Curve

We now return to the part of the preceding lecture that curvature needs. Let a spatial curve be $x^m(s)$, with tangent

$$t^p = \frac{dx^p}{ds}. \tag{1.15}$$

For a contravariant vector $v^m(s)$ defined along the curve, the covariant derivative along the curve is

$$\frac{Dv^m}{Ds} = \frac{dv^m}{ds} + \Gamma^m{}_{np} v^n \frac{dx^p}{ds}. \tag{1.16}$$

The ordinary derivative measures changes of both the vector and the local coordinate basis. The connection term removes the latter.

A geodesic is obtained by transporting its own tangent parallel to itself:

$$\frac{Dt^m}{Ds} = 0. \tag{1.17}$$

With an affine parameter this becomes

$$\boxed{\frac{d^2x^m}{ds^2} + \Gamma^m{}_{np} \frac{dx^n}{ds} \frac{dx^p}{ds} = 0}. \tag{1.18}$$

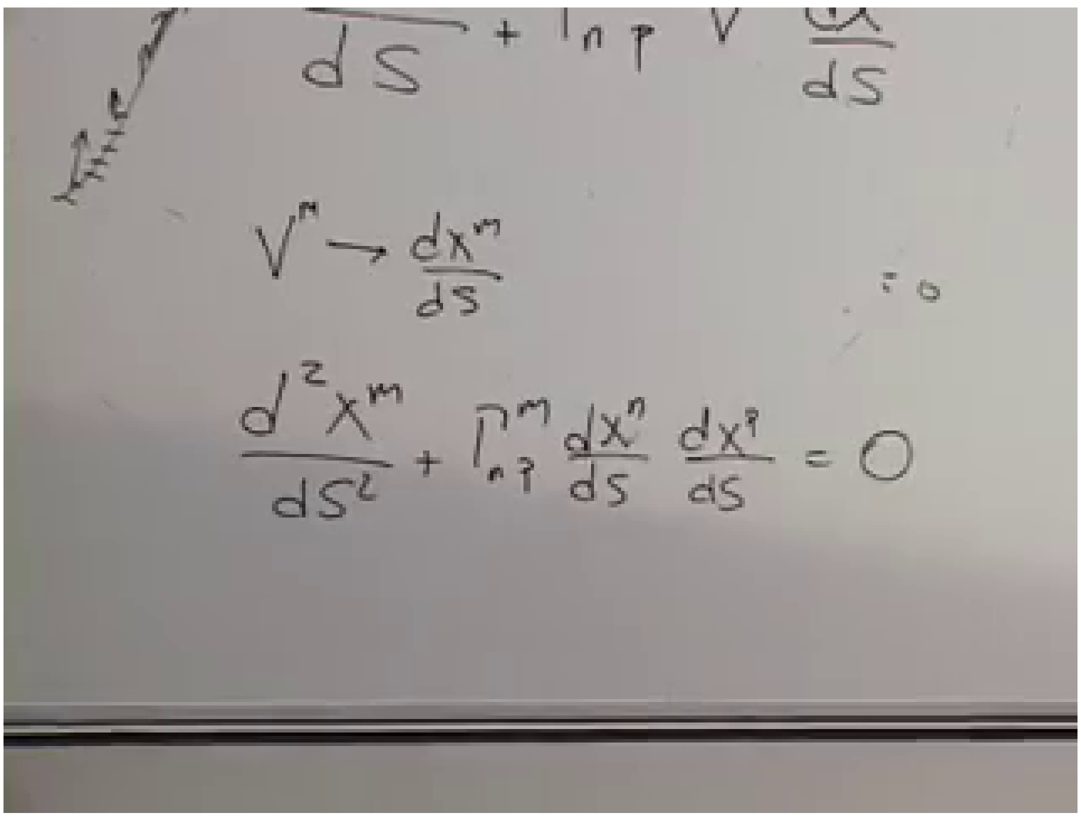

Figure 1.8: Substituting the tangent vector into the path derivative gives the geodesic equation.
Lecture frame: 00:45:48.

In a Euclidean plane the geodesics are straight lines; on a sphere they are great circles. Intrinsically, a geodesic is the curve that continues straight ahead according to the connection, whether or not it looks curved in an embedding space.

Question from the audience. Does the toy-car example mean that every vector being transported must be tangent to a shortest path? [a]

Response. No. A car that advances without steering illustrates a geodesic because its direction is its tangent. Parallel transport is more general: any vector can be transported along any prescribed curve, whether or not that curve is geodesic and whether or not the vector is tangent to it.

[a] Lecture timestamp: 00:50:30–00:51:52.

1.7 Parallel Transport of an Arbitrary Vector

Set Eq. (1.16) to zero for a general vector:

$$\frac{Dv^m}{Ds} = 0. \tag{1.19}$$

Multiplying by ds gives the infinitesimal update rule

$$\boxed{dv^m = -\Gamma^m{}_{np} v^n \, dx^p}. \tag{1.20}$$

Given v^m at one point, the connection and the next displacement determine its components one step later. Repeating this operation solves a first-order differential equation along the entire curve.

Metric compatibility guarantees that parallel transport preserves inner products. Write $u \cdot v \equiv g_{mn} u^m v^n$. If u^m and v^m are both parallel transported, then

$$\begin{aligned}\frac{D}{Ds}(u \cdot v) = \\ &(\nabla_p g_{mn}) t^p u^m v^n \\ &+ g_{mn} \frac{Du^m}{Ds} v^n \\ &+ g_{mn} u^m \frac{Dv^n}{Ds} \\ = 0.\end{aligned} \tag{1.21}$$

Lengths and mutual angles therefore remain fixed. The angle between a transported vector and the tangent of a nongeodesic curve need not remain fixed, because the tangent itself is turning.

Question from the audience. Does parallel transport mean keeping a constant angle between the vector and the tangent to the curve? [a]

Response. Not for an arbitrary curve. In a flat plane, a vector with fixed Cartesian direction is parallel transported around a circle, yet its angle to the circle's tangent changes continuously. If the curve is itself a geodesic, its tangent is also parallel transported, so the angle between the two transported vectors remains constant.

[a] Lecture timestamp: 00:52:05–00:53:40.

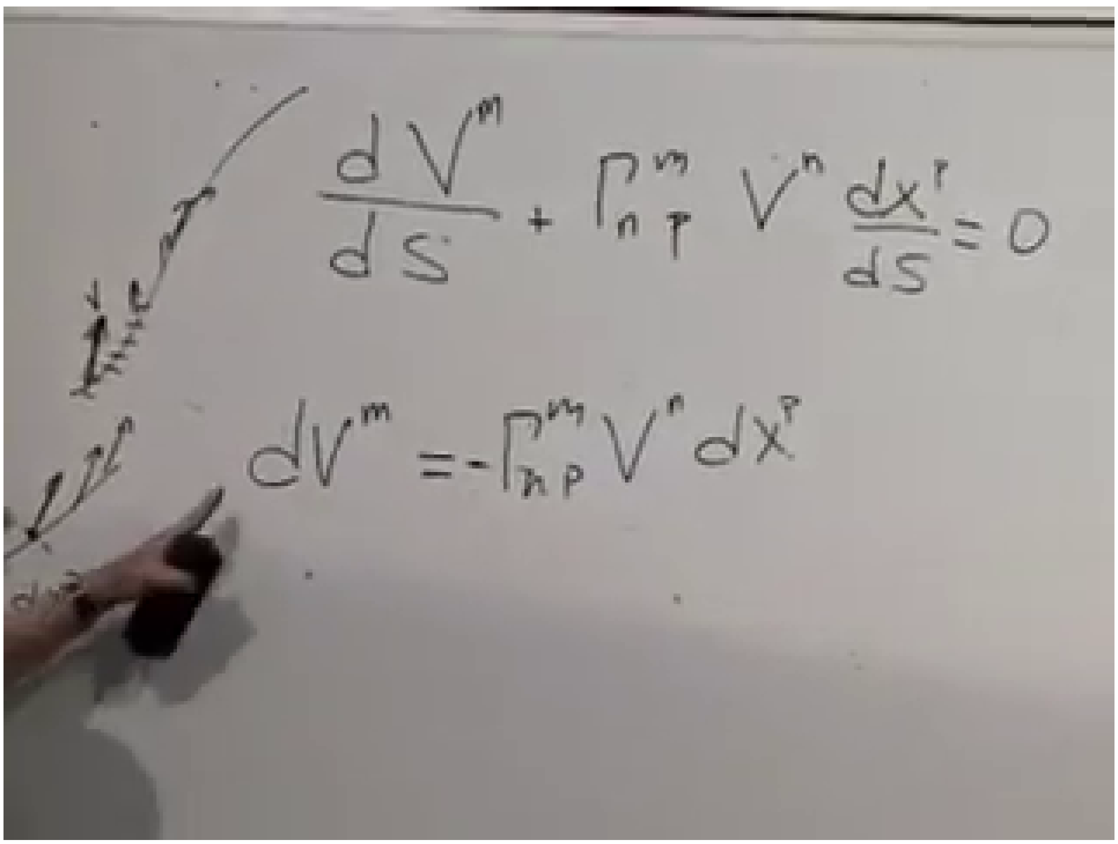

Figure 1.9: The differential equation and incremental update law for parallel transport.

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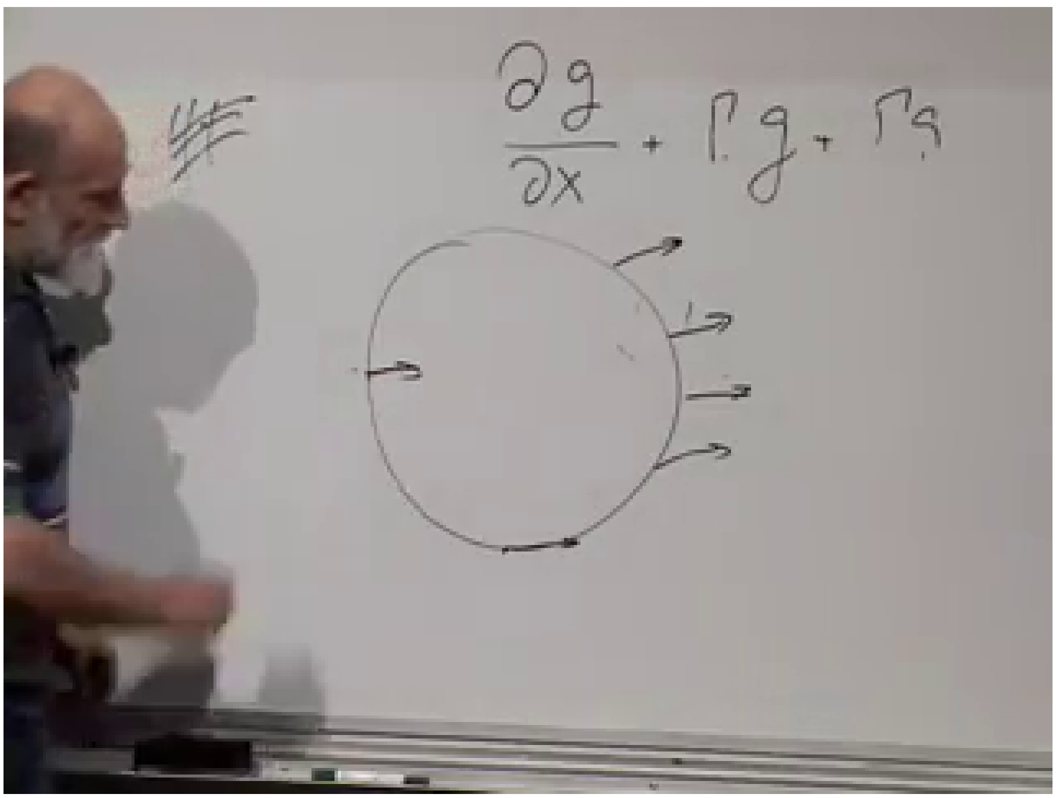

Figure 1.10: A vector kept parallel in a flat plane while moving around a circle. Its angle relative to the changing tangent is not constant.

Lecture frame: 00:52:50.

Question from the audience. Must s be arc length in the parallel-transport equation? [a]

Response. No. Equation (1.20) depends on the ordered displacements dx^p, not on the label used to parameterize them. The geodesic equation in the simple form (1.18), by contrast, requires an affine parameter.

[a]Lecture timestamp: 00:53:42–00:53:55.

A gyroscope supplies a physical picture in ordinary space. Move its center along a complicated path without applying a torque, and its axis retains its inertial direction. More abstractly, a transported orthonormal frame remains orthonormal.

Question from the audience. If parallelism implies perpendicular directions, what happens to two initially perpendicular vectors? [a]

Response. Transport both by the same Levi–Civita connection and their inner product remains zero. Three mutually perpendicular vectors likewise remain an orthonormal triad. This is the geometric content of Eq. (1.21).

[a]Lecture timestamp: 00:55:33–00:56:50.

1.8 Path Dependence and Holonomy

Now ask the decisive question. If a vector is transported from A to B along two different curves, must the final vectors agree? Equivalently, if it is transported around a closed curve, must it return unchanged?

The transport equation always refers to the chosen path. In flat Euclidean space with ordinary Cartesian connection, every contractible closed loop returns the vector to itself. On a curved surface, sufficiently small loops generally return it rotated. The resulting linear transformation of the tangent space is called the *holonomy* of the loop.

Question from the audience. After ds is canceled, why does the transport law still depend on a curve? [a]

Response. The parameter has disappeared, but dx^p has not. Those infinitesimal displacements, in their order, are the curve. A different path supplies a different sequence of dx^p and can produce a different final vector.

[a]Lecture timestamp: 00:57:44–00:58:04.

For small contractible loops, nontrivial holonomy is a local diagnosis of curvature. For large loops, topology can also matter. Keeping those two effects separate will become important in the cone, the Möbius band, and the torus examples.

Question from the audience. Does a Möbius strip have a vector-return effect of this kind? [a]

Response. Yes, but it is a global topological effect rather than ordinary local Gaussian curvature. A flat Möbius band is locally Euclidean, yet transport around its noncontractible central loop can reverse the transverse direction because the surface is nonorientable.

[a]Lecture timestamp: 01:00:44–01:01:14.

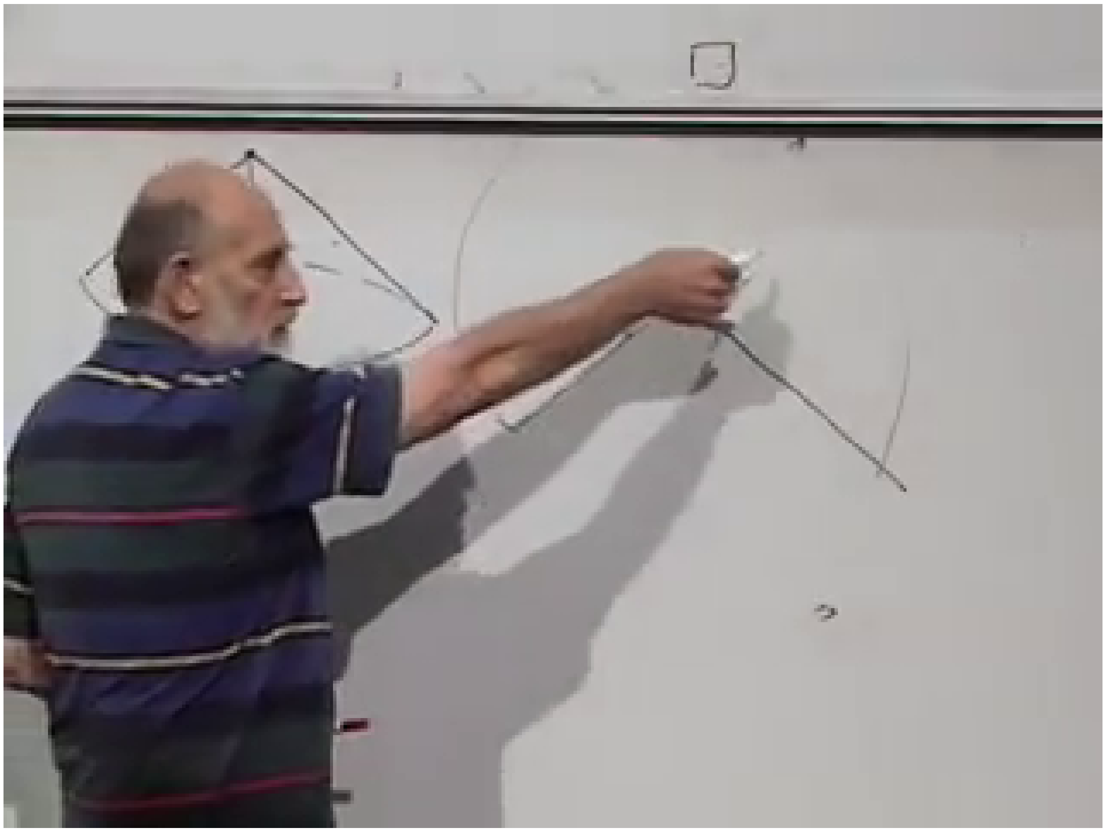

Figure 1.11: A cone and its opened representation as a flat sheet with an angular wedge removed and the two radial edges identified.

Lecture frame: 01:04:45.

1.9 The Cone: Curvature Concentrated at One Point

Cut a wedge of angle Θ from a flat sheet and identify the two cut edges. The result is a cone. When the cone is cut open again, it is a flat sector with an angular deficit Θ. Points on one cut edge are identified with corresponding points on the other.

Every point away from the apex has an ordinary flat neighborhood. A small loop that does not enclose the tip has trivial holonomy. A loop that winds once around the tip is different. On the opened sheet, keep the vector parallel in the usual Euclidean sense. When the identified edges are joined again, the final vector is rotated relative to the initial one.

Up to the sign fixed by orientation, the rotation equals the angular deficit:

$$\Delta\alpha = \Theta. \tag{1.22}$$

Any loop winding once around the apex gives the same result, while a loop that can be contracted without crossing the apex gives zero.

Smoothing the tip spreads this curvature over a small cap. A loop outside the cap detects the same total rotation but cannot tell how curvature is distributed inside. A smaller loop within the cap detects only the curvature it encloses. In the ideal sharp-cone limit,

$$\int_\Sigma K\, dA = \Theta \tag{1.23}$$

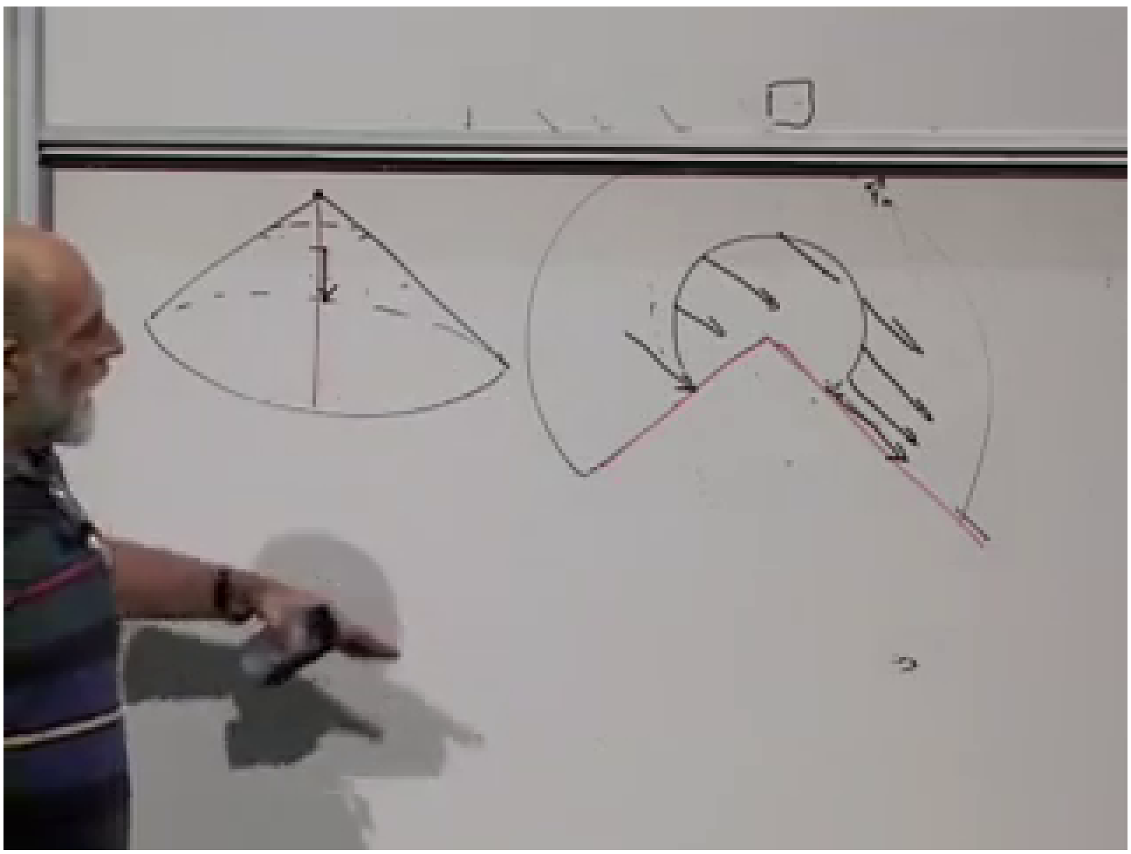

Figure 1.12: Parallel transport around the cone, shown on the opened flat sector. Joining the cut edges turns the visible mismatch into a rotation at the original point.

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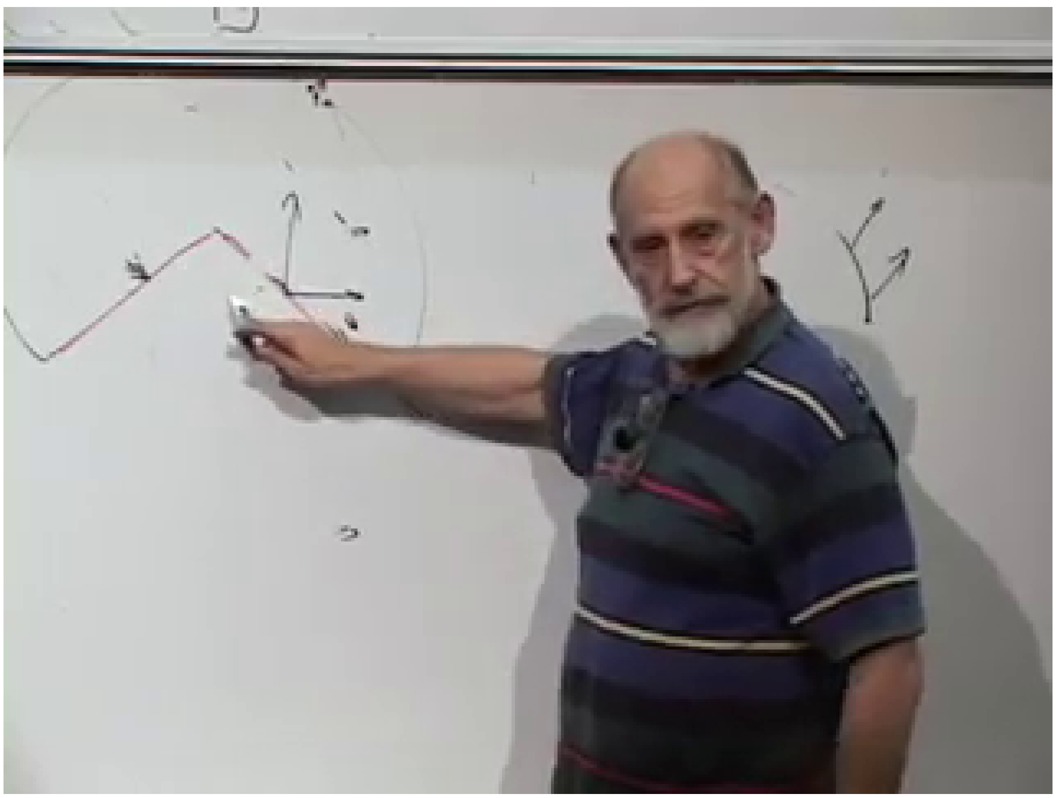

Figure 1.13: The angular deficit Θ controls the rotation acquired by a vector transported once around the cone tip.

Lecture frame: 01:10:45.

for any region Σ containing the apex. Thus the cone's curvature is concentrated there as a delta-function singularity. The lecture's phrase *infinite curvature at the tip* is this singular limiting statement, not a finite pointwise value.

Question from the audience. Can parallel transport be made more precise than keeping the vector in the same direction, and what happens along a geodesic? [a]

Response. The precise definition is $Dv^m/Ds = 0$. One may approximate the path by tiny segments and apply the incremental law on each segment. Along a geodesic the tangent obeys the same equation, so in two dimensions a transported vector keeps a constant angle with that tangent. The lecture briefly mentions twisting in higher dimensions; the present connection remains the torsion-free Levi–Civita connection.

[a] Lecture timestamp: 01:13:26–01:15:07.

Question from the audience. What do geodesics on the cone look like, especially when they cross the identified cut? [a]

Response. Open the cone along a cut that does not interrupt the segment of interest. Geodesics are straight lines on that flat sector. When the original cut is restored, such a line may look bent in the opened drawing, but its two pieces meet smoothly on the actual cone. Changing the cut changes the picture, not the intrinsic geodesic.

[a] Lecture timestamp: 01:15:07–01:18:24.

1.10 The Sphere and the Small-Loop Law

A sphere has smooth curvature everywhere. Begin with a closed line of latitude rather than a great circle. The great circles are geodesics; a line of latitude is not. Transporting a vector around it produces a net rotation.

To connect this with the cone, fit a cone tangent to the sphere along the chosen latitude. An observer confined to an infinitesimal neighborhood of that curve sees the same first-order geometry along the path. Near the north pole the tangent cone is nearly flat and its deficit is small; larger enclosed regions produce larger rotations.

For a sufficiently small oriented loop enclosing area δA,

$$\delta\alpha = K(p)\,\delta A + \text{higher-order terms.} \tag{1.24}$$

The board writes this as

$$\delta\theta = R\,a, \tag{1.25}$$

where a denotes the small area and R is the local two-dimensional curvature coefficient. We use K for Gaussian curvature so that the board symbol is not confused with the later Riemann tensor or Ricci scalar. For a sphere of radius ρ,

$$K = \frac{1}{\rho^2}; \tag{1.26}$$

the same small area therefore gives less rotation on a larger sphere.

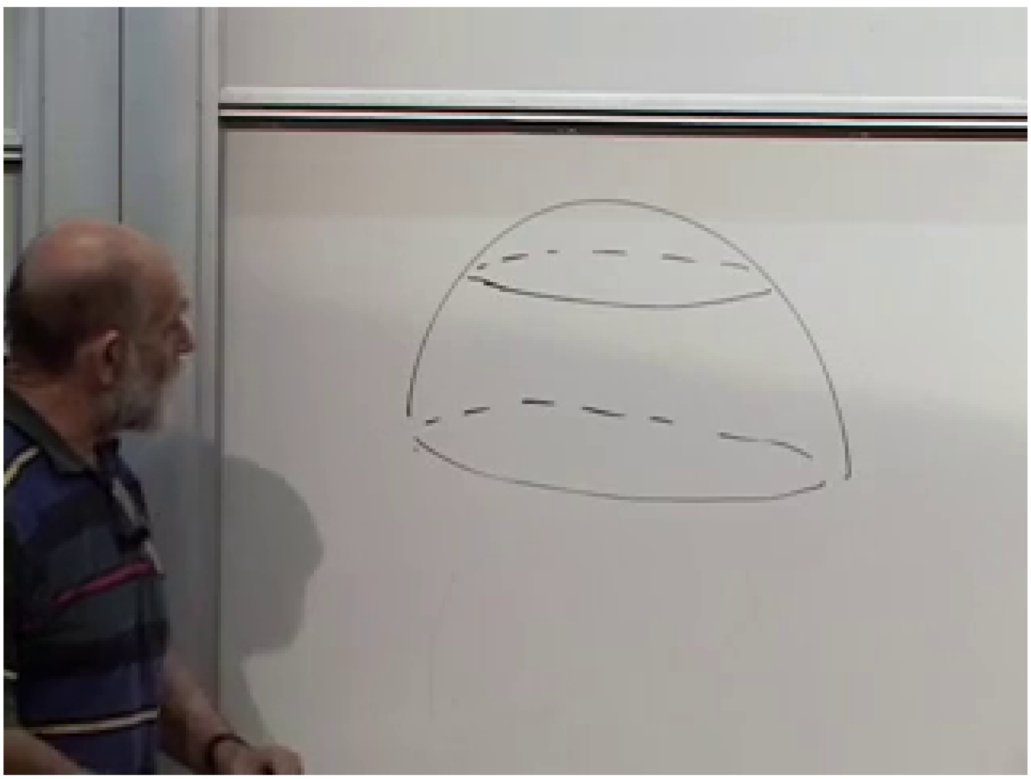

Figure 1.14: Closed latitude curves on a hemisphere. Their transport holonomy diagnoses the sphere's smooth curvature.

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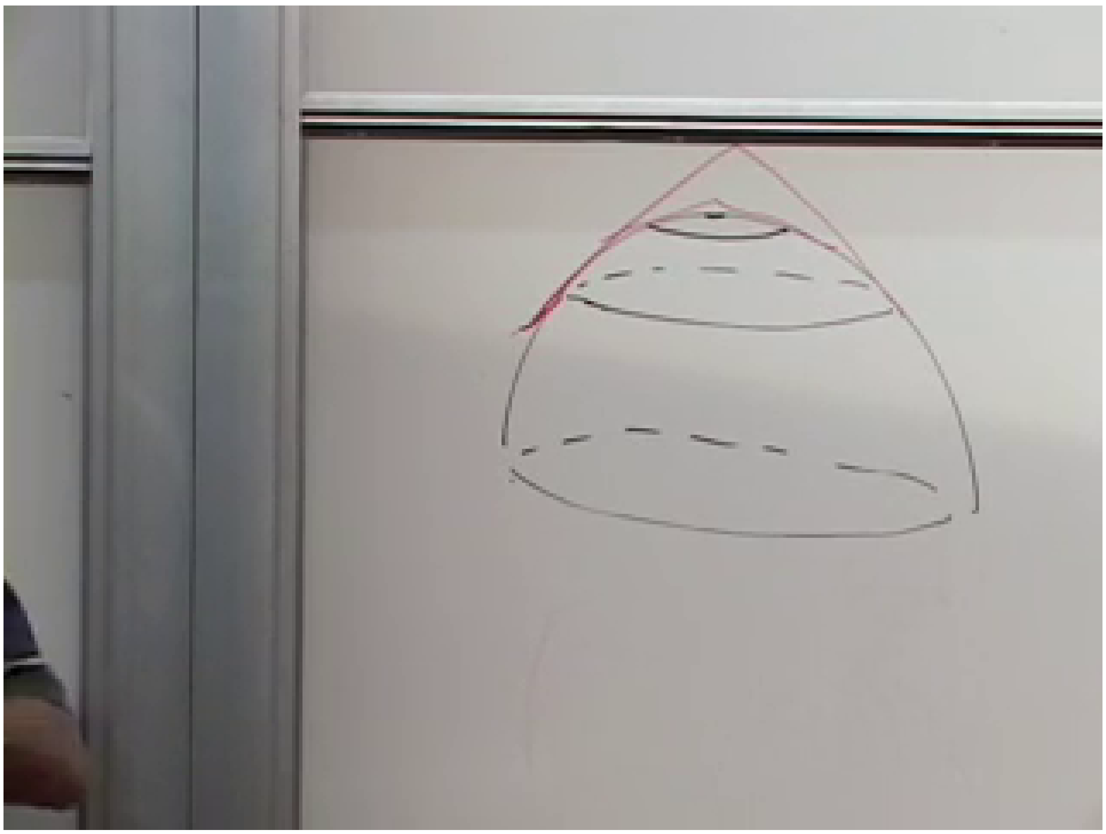

Figure 1.15: A family of latitude loops and the tangent-cone construction used to estimate their transport rotation.

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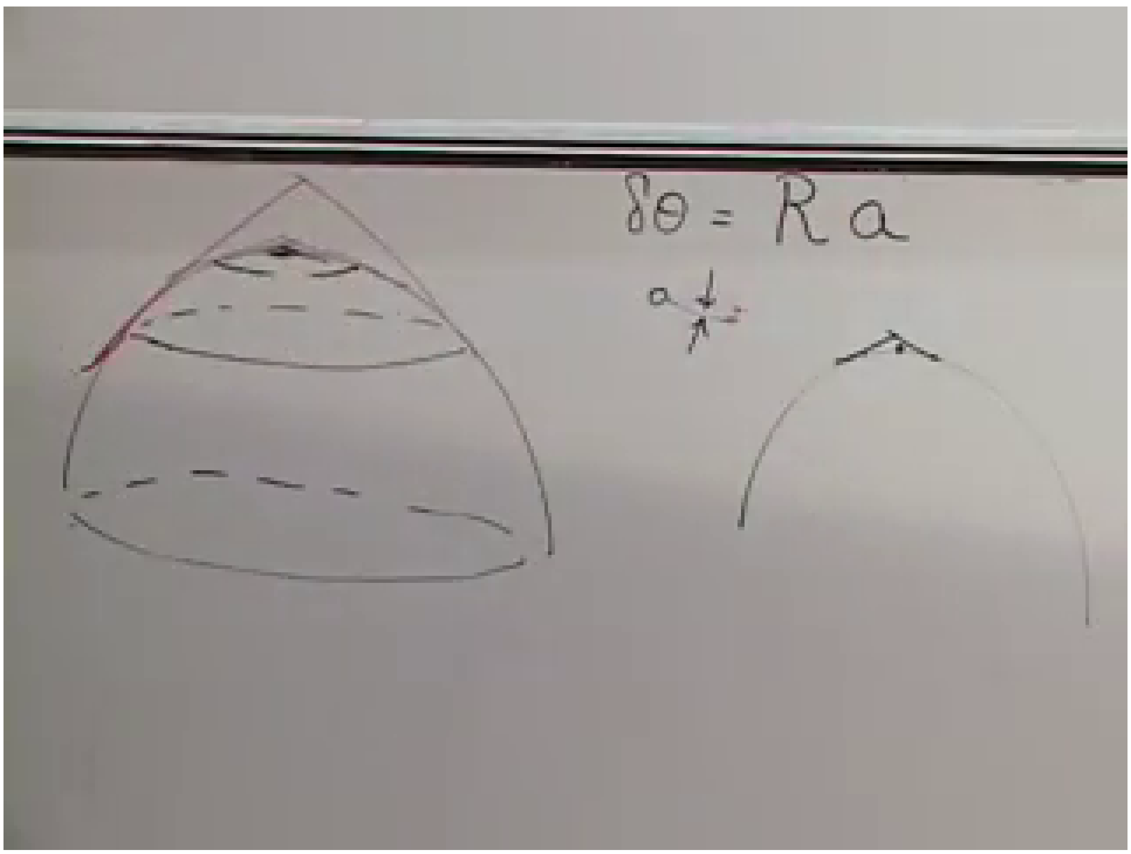

Figure 1.16: The local two-dimensional law written on the board as $\delta\theta = Ra$: holonomy angle equals curvature times a small enclosed area.

Lecture frame: 01:25:30.

Question from the audience. Must the small loop be circular, and should its area be measured on the surface or in a projection? [a]

Response. The leading term depends on the oriented surface area, not on the detailed shape of a sufficiently small loop. Surface area is the intrinsic choice. At infinitesimal scale it differs from a tangent plane projection only at higher order, which is why the distinction does not affect Eq. (1.24).

[a] Lecture timestamp: 01:24:45–01:26:51.

Question from the audience. For an embedded sphere, do the transported vectors lie on the curved surface or in the surrounding three-dimensional space? [a]

Response. A vector at p belongs to the intrinsic tangent space T_pM. In an embedding picture that tangent space is represented by the plane tangent to the sphere at p; the vector need not be the tangent of the particular curve being followed. As the base point moves, the tangent plane changes.

[a] Lecture timestamp: 01:26:52–01:28:11.

Question from the audience. What happens for a large loop such as the equator, where the small-area formula is no longer valid? [a]

Response. One integrates the curvature over the enclosed region: $\Delta\alpha = \int_\Sigma K\,dA$, with the angle understood modulo 2π and with the appropriate orientation. The equator bounds a hemisphere whose integrated curvature is 2π, so a vector can return visibly unchanged without contradicting the nonzero curvature.

[a] Lecture timestamp: 01:28:11–01:28:35.

1.11 Topology and the Torus

Finite-loop holonomy can reveal topology as well as local curvature. The flat Möbius band is nonorientable. The flat torus is different: identify opposite sides of a Euclidean rectangle by translations,

$$(x, y) \sim (x + L_x, y), \qquad (x, y) \sim (x, y + L_y). \tag{1.27}$$

The quotient is intrinsically flat even though it cannot be drawn as the usual smooth doughnut in three-dimensional Euclidean space without distorting distances.

The familiar embedded torus has positive Gaussian curvature on its outer side and negative curvature near the inner rim. Its total Gaussian curvature is zero:

$$\int_{\mathbb{T}^2} K\,dA = 0. \tag{1.28}$$

This is the torus case of the Gauss–Bonnet theorem. Zero total curvature does not mean zero curvature point by point.

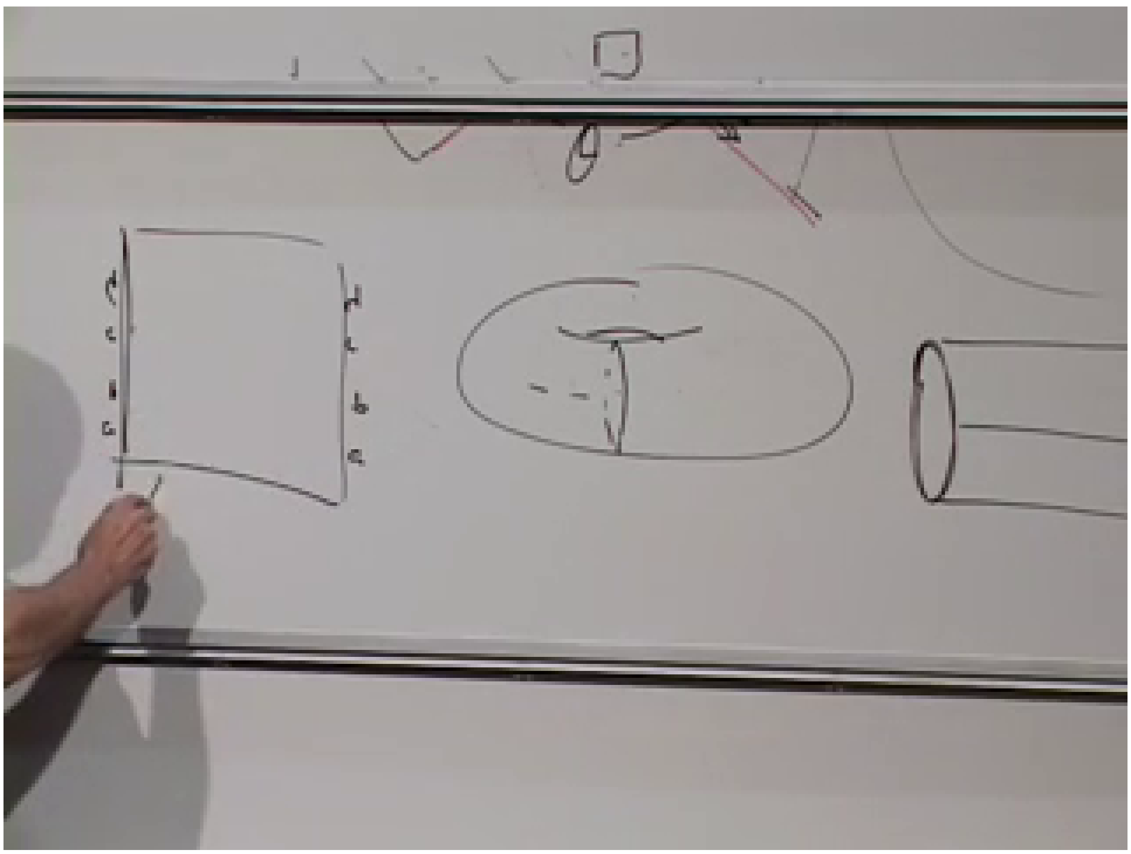

Figure 1.17: The torus as a rectangle with opposite edges identified, contrasted with cylinder and doughnut pictures. The quotient metric can be flat even though the familiar embedding is curved.

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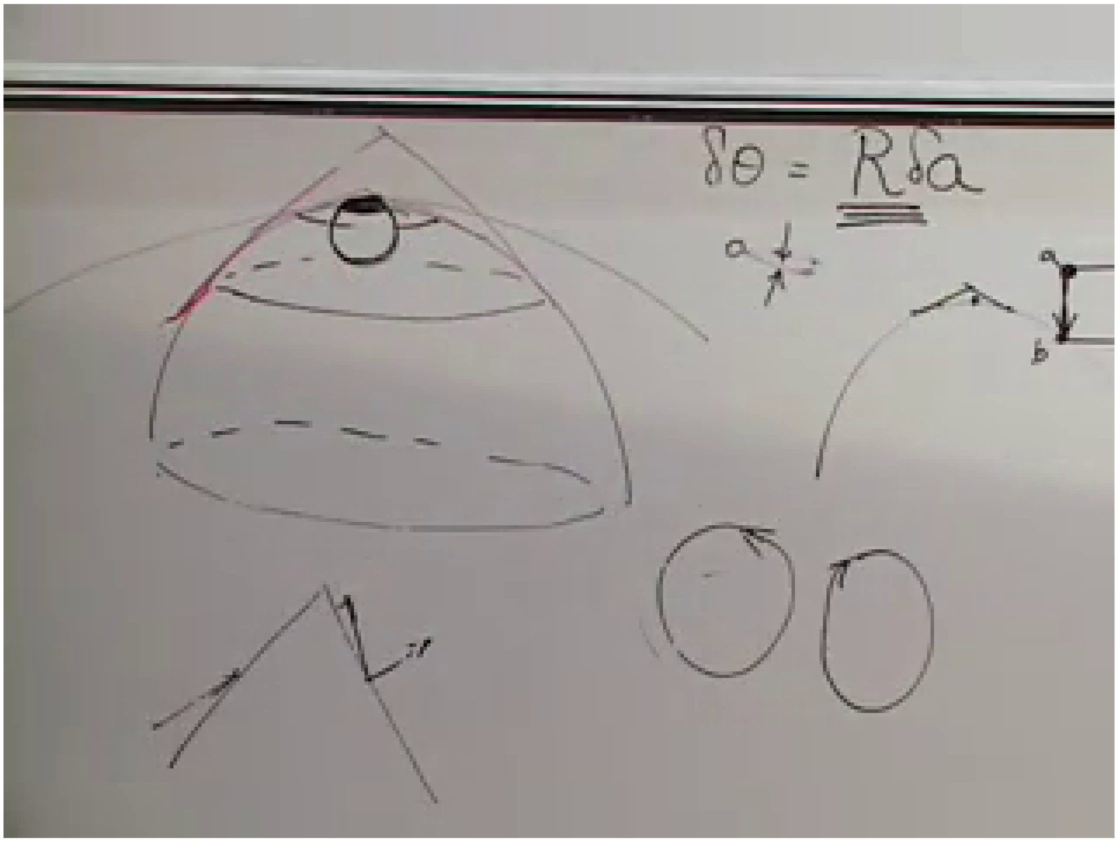

Figure 1.18: Oriented loops and vector rotations used to assign a sign to the two-dimensional curvature coefficient.

Lecture frame: 01:39:45.

Question from the audience. Can the flat rectangular torus be physically folded into an ordinary doughnut, and what curvature does a real torus have? [a]

Response. The rectangle with translated edge identifications is an abstract flat torus. A smooth isometric embedding of that flat metric does not exist in ordinary three-dimensional Euclidean space; folding paper introduces creases or stretching. It can be embedded smoothly in higher dimension. The standard doughnut in three dimensions is not flat: its outer region has positive curvature, its inner region negative curvature, and the integral cancels to zero.

[a]Lecture timestamp: 01:30:34–01:36:20.

1.12 The Sign of Curvature

Give the boundary of a small area an orientation. Call counterclockwise positive and clockwise negative. Once a convention for vector rotation is fixed, positive curvature means that the holonomy rotation and the oriented area have the same sign; negative curvature means they have opposite signs.

Removing a wedge creates positive concentrated curvature at the cone tip. The complementary construction is to cut the sheet and insert an extra wedge, producing an angular excess. A vector

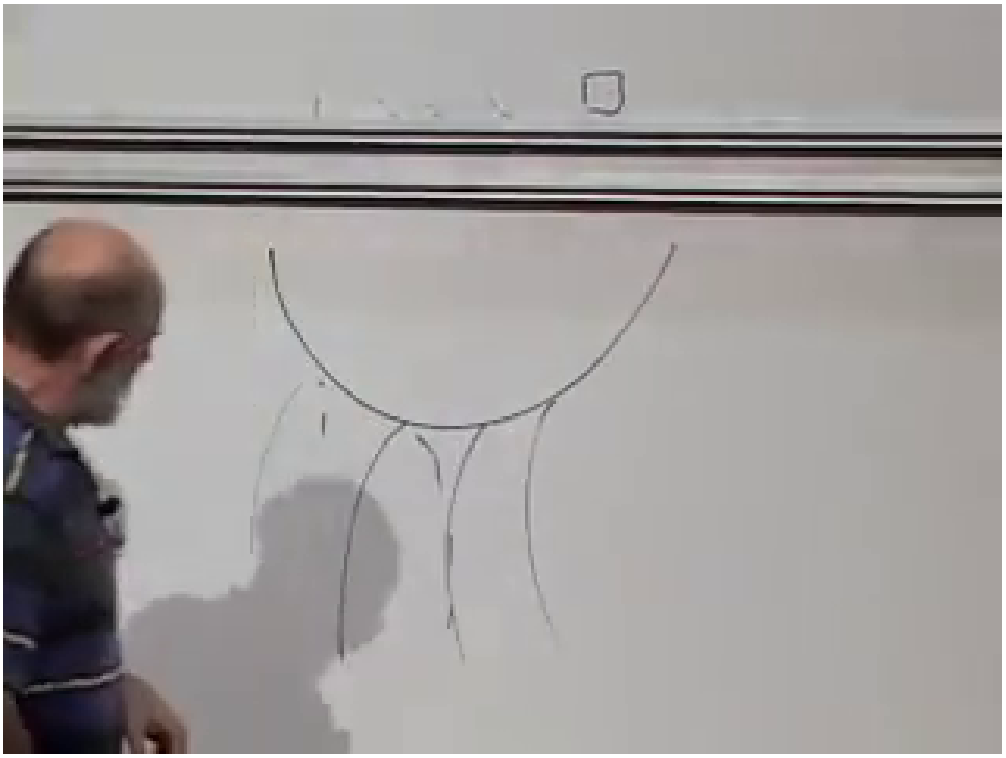

Figure 1.19: A saddle-shaped patch, the characteristic local picture of negative Gaussian curvature.
Lecture frame: 01:40:30.

transported counterclockwise then rotates in the opposite direction. This is a concentrated model of negative curvature.

Smooth negative curvature has the local shape of a saddle rather than a sphere.

The embedded torus exhibits both signs on one connected surface. Its sphere-like exterior is positively curved; the saddle-like region near the hole is negatively curved.

Question from the audience. If small loops are drawn at different places on a real torus, do they all give the same curvature? [a]

Response. No. The coefficient $K(p)$ varies over the embedded torus. It is positive outside, negative inside, and passes through zero between those regions. Only its integral over the whole torus vanishes.

[a] Lecture timestamp: 01:35:25–01:36:20.

1.13 Back to Space-Time and Gravity

Everything we have done has a space-time counterpart. Replace the positive-definite spatial metric by a Lorentzian metric, transport four-vectors with its Levi–Civita connection, and examine what happens around small loops in different coordinate planes. In more than two dimensions both the plane of the loop and the plane in which the vector rotates matter. The object encoding all of those possibilities is the Riemann curvature tensor, which is the subject of the next lecture.

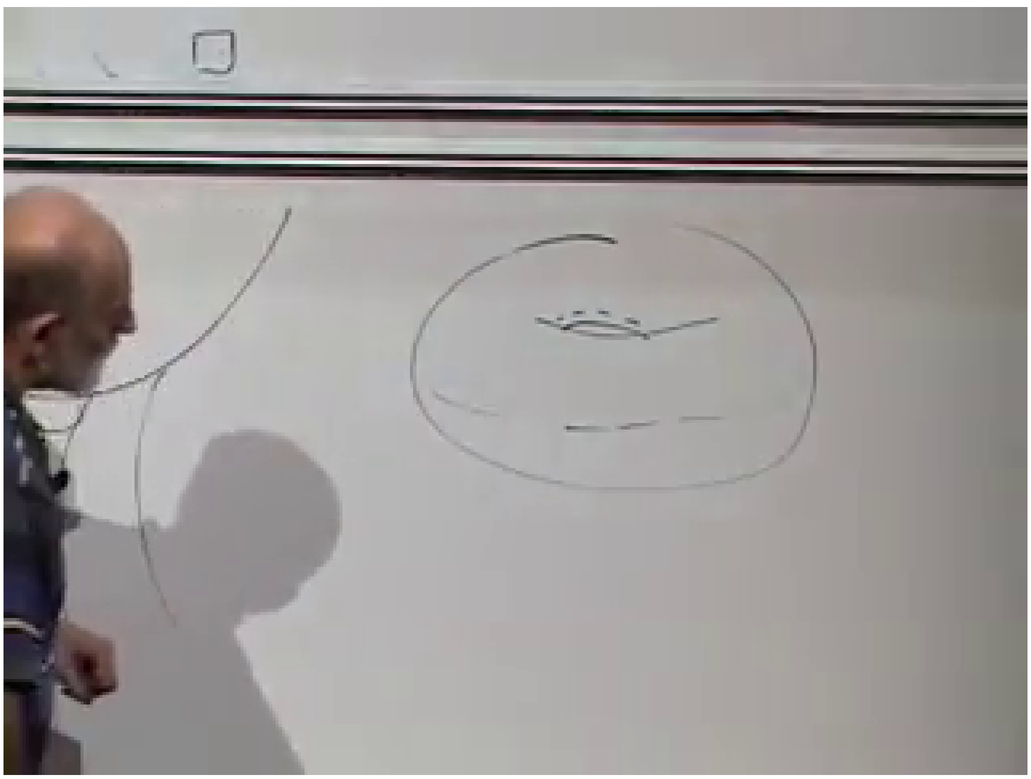

Figure 1.20: The torus cross-section used to distinguish the positively curved outer region from the negatively curved inner region.

Lecture frame: 01:43:45.

The physical program of general relativity can now be stated without yet writing its field equation:

$$\begin{gathered}\text{energy and momentum}\\ \downarrow\\ \text{space-time curvature}\\ \downarrow\\ \text{geodesic motion}\end{gathered} \tag{1.29}$$

Mass is one form of energy, and energy and momentum together source the curvature that cannot be removed by an accelerated frame throughout a region. Freely falling particles follow space-time geodesics. In the weak-field, slow-motion limit those geodesics reproduce Newtonian trajectories, with relativistic corrections. The Schwarzschild geometry will provide the central example.

Question from the audience. If a curved surface is embedded in a flat higher-dimensional space, is intrinsic parallel transport just keeping the vector parallel in the embedding space? [a]

Response. No. Keeping an ambient vector fixed would generally make it leave the surface's tangent plane. Intrinsic Levi–Civita transport keeps the vector tangent and removes only the tangent component of its change. In an embedding picture, one may differentiate in the ambient space and project back onto the changing tangent plane. The definition itself needs only the metric on the surface, not the embedding.

[a] Lecture timestamp: 01:48:54–01:51:33.

The central lesson is therefore local but powerful. At every regular event we can choose a freely falling frame in which the connection vanishes. After moving around a loop, those local frames need not fit back together. That mismatch is curvature, and in general relativity it is the invariant content of the gravitational field.